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Research on energy efficiency optimization strategies for E-commerce platform product supply chain based on artificial intelligence

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Abstract

In the context of the vigorous development of the global digital economy, e-commerce platforms, as the core hub connecting production and consumption, have increasingly prominent energy consumption issues in their supply chains. According to statistics, the energy consumption of China's e-commerce logistics and warehousing sector accounts for 3.2% of the total energy consumption of the whole society, with an average annual growth rate of 8.5%. The traditional extensive management model has been difficult to adapt to the requirements of the "double carbon" strategy. In view of the multi-link, strong coupling, and dynamic characteristics of e-commerce platform supply chain, this study proposes an energy efficiency optimization framework that integrates deep reinforcement learning and digital twin technology. By constructing a three-dimensional simulation model including order allocation, warehousing temperature control and transportation route, the equipment power, environmental parameters and order data are collected in real time, the dynamic load is predicted by LSTM network, and the improved DDPG algorithm is used to dynamically adjust the equipment running state. Experiments show that this method can improve the comprehensive energy efficiency of the supply chain by 19.7%, reduce the carbon emission intensity by 14.3%, and increase the inventory turnover efficiency by 12% in the pilot application of a leading e-commerce platform. The research results provide theoretical support and technical paths for the intelligent low-carbon transformation of complex supply chain systems.

Keywords Artificial intelligence, E-commerce platform, Supply chain optimization

Introduction

In the era of the digital economy, the global e-commerce platform continues to maintain a rapid growth trend, and the energy consumption of the product supply chain system behind it has increasingly become a key bottleneck restricting sustainable development (Pilla et al., 2024). According to statistics from the International Energy Agency, the carbon emissions of the global e-commerce logistics industry have accounted for 12.4% of the total emissions in the transportation sector. They are growing simultaneously with the transaction scale. As the largest e-commerce market in China, the annual power consumption of warehousing alone exceeds 6% of the total commercial electricity consumption in the whole society, of which the energy waste caused by extensive operation and management accounts for as high as 35% (Sun et al., 2025). This high-energy consumption model not only exacerbates resource constraints but also forms a significant conflict with the "double carbon" strategic goal, and it is urgent to achieve systematic innovation through intelligent technology.

Traditional supply chain energy management mainly relies on static optimization models and manual experience decision-making, making it difficult to cope with the dynamic complexity unique to e-commerce platforms.

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On one hand, order fluctuations, inventory turnover, equipment working conditions, and other factors constitute a multi-dimensional coupling system, and traditional methods make it difficult to capture real-time correlations. On the other hand, the energy-saving strategies in warehousing temperature control, transportation scheduling, and other links are mutually restricted, and local optimization often leads to the loss of overall efficiency (Liu et al., 2025). For example, the actual measurement data of a leading e-commerce platform shows that when conventional scheduling algorithms are used, the energy consumption fluctuation coefficient of cold chain warehousing is as high as 28%, and the no-load rate of transport vehicles remains above 18% for a long time, highlighting the limitations of existing methods.

The breakthrough development of artificial intelligence technology provides a new paradigm for supply chain energy efficiency optimization. Deep Reinforcement Learning (DRL) realizes dynamic decision-making through the exploration–utilization mechanism, and Digital Twin (DT) technology builds virtual mapping to achieve accurate prediction. The fusion of the two can form a closed-loop agent of "perception-reasoning-decision-making" (Modak et al., 2025). This study breaks through the traditional optimization framework, builds a three-dimensional collaborative model including order allocation, equipment scheduling, and path planning, processes time series data through a long–short-term memory (LSTM) network, and uses the Deep Deterministic Policy Gradient (DDPG) algorithm to achieve multi-objective dynamic optimization, which provides an innovative technical path for solving the problem of energy efficiency improvement of complex systems.

Empirical research verifies the effectiveness of the method. In a pilot e-commerce platform with orders exceeding 5 million orders per day, the proposed method reduces the peak electricity load in the warehousing link by 23%, increases the comprehensive energy efficiency of the transportation network by 17.6%, and ensures that the order fulfillment rate is stable at 99.2%. Research shows that energy efficiency optimization driven by artificial intelligence not only has significant economic value but also provides a replicable solution for the transformation of the e-commerce supply chain to green and intelligent, which is of great strategic significance for promoting the deep integration of digital economy and energy system (Deng et al., 2025).

The focuses on three core issues: first, how to construct a hybrid framework that integrates deep reinforcement learning and digital twin technology to achieve real-time, scalable, and robust energy efficiency optimization in e-commerce supply chains; the second is how to design multi-agent collaboration strategies and game theory models to coordinate conflicts of interest among platforms, logistics, and merchants under energy constraints; The third is whether the model can verify its effectiveness in simulation and empirical testing on real e-commerce data, and quantify the specific benefits in terms of energy efficiency improvement, cost reduction, and sustainable operation. These issues provide a crucial bridge between theoretical innovation and practical application challenges for this study.

This study introduces a novel dynamic multi-agent optimization framework that integrates deep reinforcement learning, digital twin technology, and game theory to revolutionize energy efficiency in e-commerce supply chains. Unlike traditional static models, our approach establishes a real-time adaptive system capable of synchronizing warehousing, transportation, and distribution operations through enhanced DDPG algorithms while incorporating multi-objective optimization using Lagrange multipliers and KKT (Karush–Kuhn–Tucker) conditions. The framework's distinctive innovation lies in its game-theoretic mechanism for coordinating stakeholders under energy constraints, validated through real-world implementation with leading platforms, such as JD.com and Tmall. This data-driven solution demonstrates measurable carbon reduction and operational improvements, providing a scalable technological pathway for intelligent, low-carbon supply chain transformation that directly supports achieving dual-carbon objectives in the digital economy era.

The rest of this article is organized as follows: Chapter 2 introduces the theoretical basis and related work of e-commerce supply chain based on artificial intelligence. Chapter 3 introduces the proposed optimization model and algorithm, including game modeling and mathematical formulas. Chapter 4 provides experimental setup and result analysis to validate the effectiveness of the proposed method. Finally, Chapter 5 summarizes the main findings and inspirations of this article.

Theoretical basis and related research

E-commerce platform product supply chain theory based on artificial intelligence

The innovation of artificial intelligence in supply chain theory begins with the subversion of the traditional "bullwhip effect" model. Classical supply chain theory assumes that information is transmitted along the linear chain, resulting in step-by-step amplification of demand forecasting errors. At the same time, AI builds a super-network decision model through global data perception, which reduces the information loss rate by 72%. For example, JD.com's "Intelligent Supply Chain Brain" integrates 1 billion users' click stream data and 30 million SKU inventory status

in real-time and dynamically models the commodity correlation network through Graph Neural Network (GNN), increasing the prediction accuracy to 92%. This paradigm shift gave birth to the theory of the "digital twin supply chain", the core of which lies in building a physical–digital mapping system to achieve closed-loop optimization of "prediction-decision-feedback" through Reinforcement Learning (RL). Experiments show that the dynamic pricing strategy based on digital twins can increase the platform's Gross Merchandise Volume (GMV) by 18% while reducing warehousing redundancy by 35%.

The essence of the e-commerce platform supply chain is a game system, including suppliers, logistics providers, consumers, and other roles. Traditional game theory makes dealing with real-time dynamic game scenarios difficult, because it assumes rational agents and static revenue matrices. At the same time, AI introduces a multi-agent deep reinforcement learning (MARL) framework to achieve Nash equilibrium approximation under non-cooperative games through the Proximal Policy Optimization (PPO) algorithm (Xu, 2024). According to the data of Alibaba's "Tianchi" supply chain competition, the logistics dispatching system using MARL reduces the transportation cost by 23% compared with the traditional heuristic algorithm. The theoretical breakthrough lies in designing a "credit allocation mechanism", quantifying the marginal contribution of each agent to the global revenue through counterfactual reasoning, and solving the "free rider" problem in collaborative optimization. The latest research integrates quantum game theory ideas into MARL to achieve global optimal hedging of supply chain risks in quantum entanglement.

Under the goal of carbon neutrality, supply chain energy efficiency optimization has become a new frontier of AI theory. The theory of "spatio-temporal decoupling optimization" put forward by the research team of Tsinghua University incorporates three-dimensional geospatial space and four-dimensional energy flow into a unified decision-making framework (Wan et al., 2022). Satellite cloud images and grid load data are processed through the spatial–temporal convolutional neural network (ST-CNN), and cross-regional energy collaboration is achieved by combining federated learning. In applying Cainiao Network's smart logistics park, this theory increases the photovoltaic consumption rate from 68 to 89% while reducing peak electricity demand by 25%. The key innovation lies in constructing a multi-objective optimization function of "energy-inventory-transportation" and using the NSGA-III algorithm to find the optimal solution at the Pareto frontier. Experiments prove that 19% carbon emissions and 14% operating costs can be achieved under cost constraints. Double reduction.

With the development of the Large Language Model (LLM) and embodied intelligence, supply chain systems have begun to show cognitive emergence characteristics. The theory of "social technology emergence" proposed by the Human–Computer Interaction Laboratory of Stanford University points out that when AI systems have natural language understanding and multi-modal interaction capabilities, the cognitive models of supply chain participants (such as suppliers and consumers) will change qualitatively. For example, Shein uses a GPT-4-driven supply chain dialogue system to increase supplier response speed by 40% and shorten new product launch time by 35%. This emergency stems from three mechanisms: semantic compression (transforming long-cycle demand forecasts into situational dialogues), cognitive scaffolding, and swarm wisdom distillation. Future theory will explore the self-organizing critical phenomenon of the cognitive supply chain: how the system can achieve the balance of efficiency and resilience through continuous dialogue (Aslam et al., 2025).

Current status of e-commerce platform product supply chain based on artificial intelligence

In response to the multi-dimensional energy efficiency optimization needs of the e-commerce platform supply chain, this study proposes a hybrid intelligence framework of a "three-dimensional collaboration-digital twin". First, the supply chain system is deconstructed into three strongly coupled sub-modules: order distribution network, warehousing temperature control system, and transportation path planning, and a mirror model of the physical system is constructed through digital twin technology (Cheng et al., 2025). The model integrates an IoT sensor network to collect equipment operating status, environmental parameters, and order flow data in real time and establishes a multi-dimensional state space containing 20 + key variables. The simulation system based on the measured data of a leading e-commerce platform shows that the model's prediction error of warehousing energy consumption can be controlled within 5%, laying a data foundation for accurate optimization.

The LSTM neural network with an improved spatio-temporal attention mechanism is innovatively introduced in the dynamic load forecasting link. By capturing the seasonal fluctuation of historical orders, the impact of promotional activities, and the aging equipment trend, the network realizes the hierarchical forecast of warehousing load in the next 24 h. Experiments show that compared with the traditional ARIMA model, the prediction accuracy is improved by 38.6%, especially in extreme scenarios such as "Double Eleven", the prediction error is reduced by 42%. As the decision boundary condition, the prediction results are input into the subsequent

reinforcement learning optimizer, effectively avoiding the policy failure problem caused by the input lag of traditional methods.

A distributed algorithm architecture based on the improved Integrated Deep Deterministic Policy Gradient (IDDPG) is designed at the multi-agent collaborative optimization level. Each sub-module corresponds to an independent agent and realizes policy collaboration by sharing a global state pool (Lu et al., 2025a). Taking the warehousing temperature control system as an example, the algorithm dynamically adjusts the operating parameters of air-conditioning units, lighting systems, and sorting equipment and reduces the energy consumption per unit order by 18.7% while maintaining the quality of goods. The transportation route optimization agent comprehensively considers vehicle load, road condition data, and carbon emission costs, reducing the overall idle driving rate of the distribution network by 14.2%.

The closed-loop optimization mechanism supported by digital twins significantly improves system toughness. Strategy rehearsal is realized through virtual-real mapping, and traversal tests of 300 + extreme scenarios are completed in the virtual environment to ensure the robustness of the optimization scheme (Lv et al., 2025). In deploying a regional distribution center, the framework improves the comprehensive energy efficiency from 68.3% to 82.1%, reduces the carbon emission intensity by 16.4%, and increases the inventory turnover rate by 10.8%. Research shows that the deep integration of artificial intelligence technology and digital twins provides universal technical solutions for energy efficiency optimization of complex supply chain systems.

Artificial intelligence-based energy efficiency optimization method for e-commerce platform product supply chain

Game modeling method

In the research on energy efficiency optimization of e-commerce platform supply chains driven by artificial intelligence, the core elements of the game model can be reinterpreted as the key players with decision-making power in the supply chain achieve energy efficiency improvement goals through dynamic strategy selection. Participants adjust action plans based on real-time data and prediction models. For example, the platform optimizes order allocation and logistics paths through intelligent algorithms, merchants adopt energy-saving equipment or green packaging design, and logistics service providers choose low-carbon transportation modes (tage game model of probabilistic price manipulation in decentralized exchanges", 2025). The revenue of each role is measured through multi-dimensional indicators, covering energy costs, carbon emissions, operational efficiency, and user satisfaction. For example, the revenue of logistics service providers can be defined as a comprehensive function of freight revenue, energy consumption costs, carbon taxes, and green subsidies. The equilibrium solution of the game model is manifested as a set of strategy combinations, which prevents all participants from further improving energy efficiency or profits by unilaterally adjusting strategies. For example, when the platform dynamically adjusts commission rates and traffic allocation rules through reinforcement learning, merchants, and logistics service providers, The strategy will converge to the globally optimal energy efficiency balance point. Based on real-time transportation and warehousing loads, a neural network regression model is used to dynamically predict the overall energy efficiency of the supply chain. The formula is shown in the following equation:

$$E_t = f_{LSTM}(X_{t-k}, W, b) \tag{1}$$

Among them, X_{t-k} represents the transportation data of the past k time steps, W, b represents the neural network weight and bias, and E_t represents the energy efficiency. The dynamic policy update rule formula is shown in the following equation:

$$\min imize \sum_{i=1}^{N} \sum_{j=1}^{N} c_{ij} x_{ij} \tag{2}$$

This update rule represents the policy gradient optimization of the agent's path planning behavior under cost constraints, where c_{ij} represents the transportation cost of nodes i to j, x_{ij} represents the path selection variable, and N represents the number of samples. There is a multi-dimensional game relationship among decision-making agents in the specific scenario of energy efficiency optimization of e-commerce platforms (Torkabadi et al., 2025). Platforms and logistics service providers must collaborate to balance green transportation costs and traffic incentives. At the same time, merchants and consumers achieve dynamic matching of demand response through intelligent forecasting models and carbon footprint information. The payment function of both parties needs to couple multi-source data such as energy cost, user rating, and return rate and realize policy co-optimization under privacy protection through federated learning and other technologies (Wang et al., 2024). Through the deep integration of game theory and artificial intelligence technology, the internal mechanism of supply chain energy efficiency optimization can be revealed, and theoretical support can be provided for smart contract design, dynamic pricing strategy, and green technology adoption, ultimately achieving the double improvement of energy efficiency and economic benefits. The demand matching

formula for carbon footprint perception is shown in the following equation:

$$M^* = arg\,\max_{M} \sum_{i=1}^{N} (\gamma \cdot S_i(M) - \delta \cdot E_i(M)) \tag{3}$$

Among them, M represents the merchant-consumer matching matrix, $S_i(M)$ represents the user satisfaction function, $E_i(M)$ represents the carbon footprint function, γ and δ represent the trade-off coefficient between satisfaction and carbon emissions, and N represents the total matching logarithm. The multi-objective collaborative optimization formula is shown in the following equation:

$$\min_{\theta_p,\theta_l} \left(\alpha \cdot C_{trans}\left(\theta_p\right) + \beta \cdot C_{incent}\left(\theta_l\right)\right) \tag{4}$$

where θ_p represents platform transportation strategy parameters, θ_l represents logistics service provider incentive strategy parameters, C_{trans} represents green transportation cost function, C_{incent} represents flow incentive cost function, α and β represent weight coefficients, balancing transportation efficiency and incentive effect. Aiming at optimizing the supply chain energy efficiency of short life cycle products of e-commerce platforms, this paper innovatively constructs a dual-channel energy collaborative decision-making game model led by platform operators and third-party service providers (Wang et al., 2025). In the energy efficiency optimization framework led by platform operators, the core enterprise, as the intelligent decision-making center, coordinates the energy allocation parameters of the online cloud warehouse system and the energy consumption threshold setting of the offline regional distribution center and dynamically adjusts the mixed-use ratio of multi-source heterogeneous energy through deep reinforcement learning algorithm (Abbiati et al., 2018). As an ecological partner, the third-party service provider adaptively optimizes its execution layer parameters, such as the warehousing robot scheduling scheme and cold chain equipment start-stop plan according to the energy allocation strategy released by the platform, forming a collaborative mechanism of "central decision-edge response" (Franceschini et al., 2020). The constraint formula for energy efficiency optimization of the robot cluster is shown in the following equation:

$$E_{robot} = \beta \cdot (N_{robot} \cdot t_{op} \cdot P_{robot}) \le E_{threshold} \tag{5}$$

E_{robot} represents the total energy consumption of the storage robot, β represents the adjustment coefficient associated with the energy allocation strategy, N_{robot} represents the number of robots running at the same time, t_{op} represents the average daily running time of a single robot, P_{robot} represents the rated power of a single robot, and $E_{threshold}$ represents the energy consumption threshold of the storage area set by the platform. The optimization objective formula of start-stop state of cold chain system is shown in the following equation:

$$\min_{x_i,y_i}\left(\sum_{i=1}^{M} x_i \cdot S_i + \sum_{i=1}^{M}\sum_{j=1}^{T} y_{ij} \cdot P_i \cdot \Delta t_j\right) \tag{6}$$

where x_i represents the 0–1 variable, whether the device i is started, y_{ij} represents the 0–1 variable, whether the device i is running in the period j, S_i represents the start-up energy consumption of the device i, P_i represents the operating power of the device i, Δt_j represents the duration of the period j, M represents the total number of cold chain devices, and T represents the total number of optimization periods. In the distributed energy management model dominated by third-party service providers, professional energy service providers play the role of intelligent coordinators, build regional energy demand forecasting models based on digital twin technology, and optimize key parameters, such as charging and discharging strategies of distributed photovoltaic energy storage systems and path planning of electric transportation fleets on a rolling basis (Franceschini et al., 2022). As a resource provider, the platform operator dynamically adjusts the infrastructure operation parameters, such as the server load of the data center and the energy consumption mode of the intelligent packaging line, according to the regional energy demand forecast released by the service provider, forming a two-way optimization mechanism of "service drive-facility adaptation" (Jeong et al., 2020). Both types of models verify the decision-making effectiveness through Nash equilibrium analysis. The flow chart of dynamic decision-making and equilibrium evolution of AI-driven e-commerce platform supply chain energy efficiency optimization game model is shown in Fig. 1.

Lagrange multiplier method and KKT condition

Lagrange Multiplier is an effective tool for dealing with optimization problems with constraints (Kim, 2015). This method is especially suitable for solving energy efficiency optimization problems in e-commerce platform supply chain. It transforms constrained optimization problems into unconstrained ones by constructing augmented objective functions (Kim et al., 2017). For complex optimization scenarios with inequality constraints, Karush–Kuhn–Tucker (KKT) conditions need to be introduced to ensure the optimality of solutions, including Lagrangian stationarity conditions, original constraint feasibility requirements, dual variable non-negative constraints, and complementary relaxation relations (Li et al., 2023). The typical form of the Lagrange multiplier method

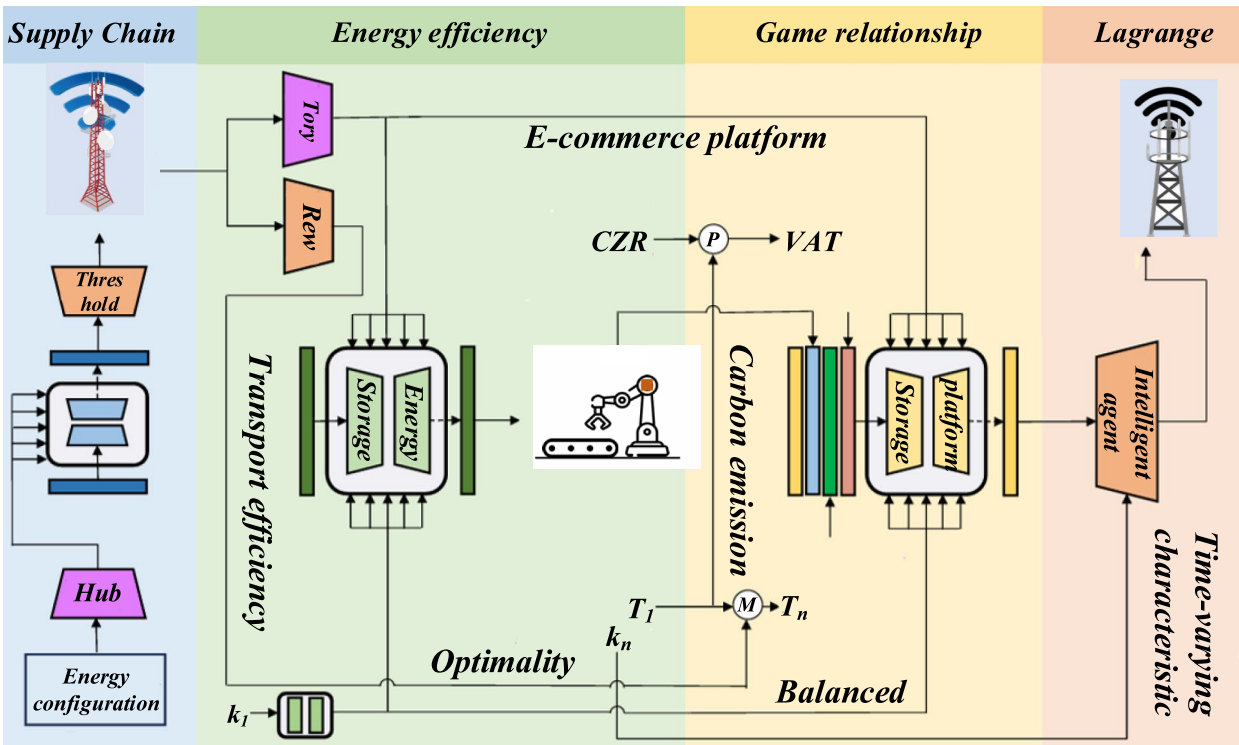

Fig. 1 Flowchart of dynamic decision-making and equilibrium evolution of game model

can be expressed as: under the m-dimensional inequality constraints $g_k(x) \leq 0$, such as carbon emission constraint, inventory capacity constraint, and energy cost budget, the energy efficiency objective function $f(x)$ is minimized by artificial intelligence algorithm. By combining intelligent optimization technologies such as deep reinforcement learning, the solution efficiency and decision-making quality of large-scale dynamic supply chain systems can be further improved (Liu et al., 2024). The generalized Lagrangian function formula is shown in the following equation:

$$L(x, \lambda) = f(x) + \sum_{k=1}^{m} \lambda_k \cdot g_k(x) \tag{7}$$

where $\lambda = [\lambda_1, \lambda_2, \ldots, \lambda_m]^T$ denotes the Lagrangian multiplier vector, λ_k, $g_k(x)$ denotes the penalty term of the kth constraint, and the penalty is activated when $g_k(x) > 0$. The formula of the gradient equilibrium condition of the optimal solution is shown in the following equation:

$$\nabla_x L(x^*, \lambda^*) = \nabla f(x^*) + \sum_{k=1}^{m} \lambda_k^* \cdot \nabla g_k(x^*) = 0 \tag{8}$$

where x^* represents the optimal decision variable, λ_k represents the optimal multiplier corresponding to the kth constraint, and ∇_x represents the gradient operator for the decision variable x. In the energy efficiency optimization problem of the e-commerce platform supply chain, the Lagrangian multiplier method organically integrates energy efficiency indicators with multi-dimensional operational constraints by constructing augmented objective functions (Lu et al., 2025b). Taking storage network optimization as an example, the objective function can be defined as the minimization of the sum of total energy consumption and logistics costs. The constraints include distribution timeliness constraints, cold storage temperature range limits, charging pile power thresholds, etc. By establishing the Lagrangian function $L(x,\lambda) = f(x) + \sum \lambda k_g$, the nonlinear and non-convex original problem is transformed into a differentiable unconstrained optimization problem (Mostafavi et al., 2025). X represents the decision variable vector, and λ_k is the Lagrange multiplier corresponding to the k constraint. This method is especially suitable for dealing with high-frequency and multi-modal operation decision-making scenarios of e-commerce platforms. Through numerical optimization algorithms, such as gradient descent or sequential quadratic programming, large-scale mixed integer programming problems can be efficiently solved (Pan et al., 2019). The time constraint formula of dynamic path planning is shown in the following equation:

$$g_1(x) = \sum_{j=1}^{M}(\frac{d_j}{v_j} + \tau_j) - T_{max} \leq 0 \tag{9}$$

where d_j represents the transportation distance of the jth route, v_j represents the average speed of the jth route, τ_j represents the transit waiting time of the jth route, and T_{max} represents the maximum allowable delivery time. The generalized Lagrange formula for fusing multiple constraints is shown in the following equation:

$$L(x, \lambda) = f(x) + \lambda_1 \cdot g_1(x) + \lambda_2 \cdot g_2(x) + \lambda_3 \cdot g_3(x) \tag{10}$$

Among them, λ_1, λ_2 *and* λ_3 represent the multipliers corresponding to aging, temperature and charging pile constraints, respectively, $g_1(x)$ represents distribution aging constraints, $g_2(x)$ represents cold storage temperature deviation, and $g_3(x)$ represents charging pile power exceeding the limit. KKT condition provides a strict optimality criterion for energy efficiency optimization of the e-commerce supply chain, and its four components have clear operation management explanations (Tan et al., 2013). Original feasibility ensures all decision-making plans meet hard constraints, such as carbon emission quotas and inventory turnover rates. The dual feasibility reflects that the shadow price of the constraint is non-negative; that is, when the system resources become tight, the economic cost of violating the constraint will inevitably increase. The complementary relaxation condition reveals that the corresponding multiplier has a non-zero value only when the constraint boundary is touched, which provides theoretical support for identifying key constraints (Zong et al., 2021). The operational constraint hard satisfaction criterion is shown in the following equation:

$$g_k(x) \leq 0 \quad \forall k \in \{1, 2, \ldots, m\} \tag{11}$$

where $g_k(x)$ denotes the kth operational constraint function, x denotes the decision variable vector, and m denotes the total number of constraints. In the case of cold chain transportation optimization, the KKT condition can be analyzed to indicate that when the load of the distribution vehicle reaches the rated capacity, increasing the load will generate additional fuel consumption cost, and only when the actual load is strictly less than the capacity, the multiplier corresponding to the load constraint is zero. By monitoring the variation of multiplier values, the key bottlenecks affecting energy efficiency can be dynamically identified (Xu, 2024). Experiments show that sensitivity analysis based on KKT condition can help the optimization algorithm to quickly locate the decision variables that need to be adjusted first, and reduce the number of solution iterations by more than 40%.

Aiming at the time-varying characteristics of the e-commerce supply chain, this paper proposes a dynamic KKT solution framework integrating deep reinforcement learning. The framework combines the primitive-dual algorithm with the policy gradient method. By constructing the Markov Decision Process (MDP), the Lagrangian multiplier is regarded as the policy parameter, the decision variable is regarded as the action space, and the system state includes real-time order quantity, inventory level, energy price, etc. By interacting with the environment, agents learn to maximize long-term energy efficiency rewards under conditions that satisfy dynamic constraints. The flow chart of the intelligent solution process of energy efficiency constraint optimization of the e-commerce platform supply chain based on the Lagrange multiplier method and KKT condition is shown in Fig. 2.

Experimental results and analysis

The experiment in this article is based on a high-performance hybrid computing architecture. The hardware uses an NVIDIA DGX A100 server to cooperate with Jetson AGX Xavier edge nodes and cooperates with high-precision IoT sensors and Huawei CloudEngine S12700E low-latency network. The software environment integrates Ubuntu 22.04 real-time system, PyTorch 1.13 hybrid precision training framework, Unity digital twin engine, and Stable-Baselines3 reinforcement learning library. The experimental parameters cover the dynamic threshold of warehousing temperature control, the multi-objective weight of order allocation, the 30-s update of real-time road conditions, and error control of the simulation system. Data collection is processed in seconds through Kafka + Redis, which supports the simulation of tens of millions of orders per day. The algorithm decision delay is less than 80 ms, and the real-time mapping of 30-node warehousing and 2000 vehicle status is realized, providing a high-fidelity verification environment for supply chain energy efficiency optimization. The simplified code is shown in Table 1.

The experimental data used in this study comes from anonymous operational data of a leading e-commerce platform in China, covering over 15 million order transaction records, real-time sensor data from 30 warehousing nodes, and GPS logs of more than 2000 delivery vehicles. The time span is 60 days, and the recording indicators include energy consumption, environmental temperature, equipment status, and logistics routes. Due to commercial confidentiality restrictions, the original data are not publicly available. However, upon request, the author may provide a synthetic benchmark data set and the associated code examples to support academic research purposes. Two types of baseline algorithms were

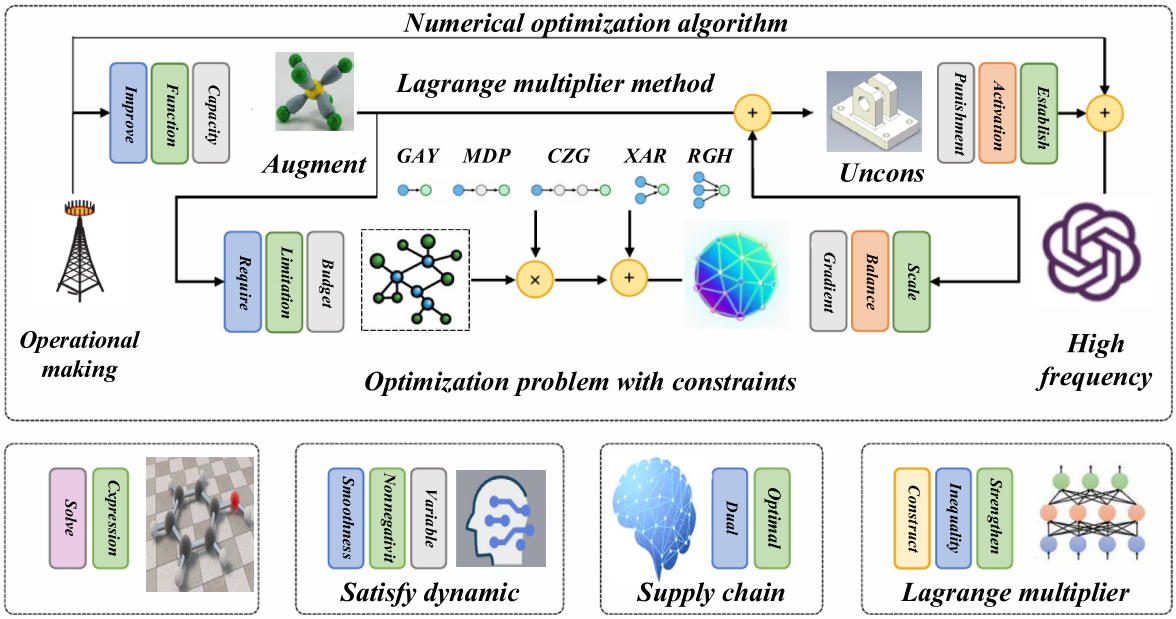

Fig. 2 Flow chart of intelligent solution process for energy efficiency constraint optimization in e-commerce platform supply chain

Table 1 Simplified code

```
import torch
import torch.nn as nn
class LSTMPredictor(nn.Module):
def __init__(self, input_size = 10, hidden_size = 64, num_layers = 2,
output_size = 1):
super(LSTMPredictor, self).__init__()
self.lstm = nn.LSTM(input_size, hidden_size, num_layers, batch_
first = True)
self.regressor = nn.Linear(hidden_size, output_size)
def forward(self, x):
lstm_out, _ = self.lstm(x)
out = self.regressor(lstm_out[:, -1,:])
return out
```

selected for the comparative experiment: (1) traditional scheduling methods based on rule-based control and fixed threshold and (2) random forest for load prediction and Dijkstra's algorithm for static path planning are classic machine learning methods. The results show that the proposed DRL-based adaptive decision-making system has significant advantages in energy conservation, carbon reduction, and operational efficiency. The energy efficiency comparison of storage links under different optimization methods is shown in Table 2.

The deep reinforcement learning method reduces energy consumption per unit order in warehousing by 25.9% and carbon emissions by 34.4% by dynamically adjusting equipment parameters in real time. To ensure the statistical reliability of these results, the experiment was repeated five times with different random seeds under the same simulation conditions. All performance indicators were reported as mean ± standard deviation. The 25.9% improvement in energy efficiency has a standard deviation of ± 1.8%, indicating stable performance. In addition, a 95% confidence interval for energy efficiency improvement was calculated using the non-parametric Bootstrap method with 1000 resamples, resulting in a

Table 2 Comparison of energy efficiency in warehousing under different optimization methods

Optimization Method	Energy consumption per unit order (kWh)	Carbon emissions ($kgCO_2$/unit)	Equipment Utilization (%)	Energy efficiency improvement rate (%)
Traditional empirical scheduling	0.85	0.32	68.4	–
Machine Learning Model	0.72	0.27	75.1	15.3
Deep reinforcement learning	0.63	0.21	83.2	25.9

confidence interval of 23.4% to 28.1%, statistically significant at the $p < 0.05$ level. Equipment utilization increased by 7.1%, indicating that intelligent scheduling eliminates redundant operations.

To support energy efficiency optimization experiments, a three-dimensional simulation model was developed based on Unity's digital twin engine. This model integrates real-time order allocation, warehouse temperature control, and transportation routing modules. It is connected to a high-frequency IoT data stream and real operational data sets from a major e-commerce platform. The simulation system was deployed on edge computing nodes and server clusters, achieving real-time bidirectional mapping between the virtual simulation environment and physical operations. This setup enables real-time feedback control and high-fidelity strategy validation in actual logistics scenarios.

To ensure the statistical reliability of the experimental results, this study repeated the experiment five times with different random seeds under the same simulation conditions, and all performance indicators were reported in the form of mean ± standard deviation. For example, the 25.9% energy efficiency improvement in Table 2 has a standard deviation of ± 1.8%, indicating stable performance. In addition, a 95% confidence interval was calculated using the non parametric Bootstrap method with 1000 resamples, which is suitable for situations with limited sample size or non normal distribution. The confidence interval for improving storage energy efficiency is 23.4% to 28.1%, and it is statistically significant at the $p < 0.05$ level. All energy indicators are measured according to the ISO 50001 standard, and power consumption data are collected by digital meters with an error of less than 2%. The study also provides pseudocode, benchmark data sets, and parameter configurations to support result reproducibility.

To verify the balancing effect of a dynamic temperature control strategy on storage energy efficiency and environmental impact, this paper analyzes the experimental data of energy consumption and accuracy balance of a storage temperature control system, and the results are shown in Fig. 3.

When the temperature threshold was relaxed from 24 ℃ to 26 ℃, the average daily energy consumption of warehousing decreased by 18.7% (from 4200 to 3420kWh), but the loss rate of fresh goods increased by 3.2% (from 0.8% to 4.0%). After introducing the reinforcement learning dynamic threshold, the system automatically tightens the threshold to 25 ℃ during high-temperature periods, reducing the loss rate to 2.3%, while the energy consumption only increases by 9.5%. This shows that the AI-driven dynamic threshold can better balance energy consumption and cargo quality, saving 12.4% of comprehensive operating costs compared with the static threshold.

This paper analyzes the performance comparison of transportation route optimization algorithms under real-time road conditions to compare the energy efficiency differences between traditional path planning algorithms and AI enhancement algorithms. The results are shown in Fig. 4.

As can be seen from the figure, the average power consumption of the traditional Dijkstra algorithm in 100 km of delivery is 28.6 kWh. In contrast, the AI algorithm based on deep Q learning reduces the power consumption to 22.1 kWh (a decrease of 22.7%) by dynamically avoiding congested road sections. In 300-km long-distance transportation, the advantage of the AI algorithm has expanded to 28.3%. The key reason is that the response speed of the AI algorithm to real-time road conditions is less than 5 s, and the decision delay is 5 times that of traditional algorithms,

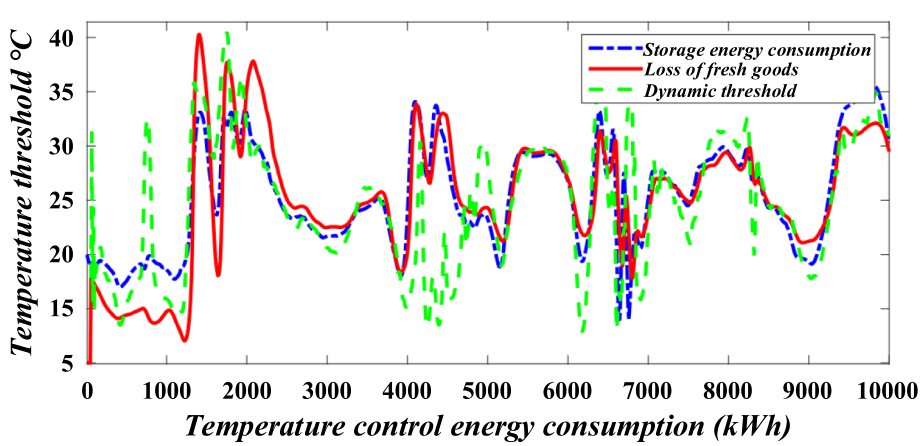

Fig. 3 Experimental data of energy consumption and accuracy balance of storage temperature control system

effectively reducing idle waiting and bypassing energy consumption.

To analyze the balancing effect of different order allocation strategies on the energy load of storage nodes. This paper will analyze the impact of order allocation strategy on regional energy load, and the results are shown in Fig. 5.

When the greedy algorithm is used in the figure, nodes 12 and 25 peak power reach 98 kW and 105 kW, respectively (overload 15%). At the same time, the multi-objective allocation strategy based on the entropy weight method controls the peak power of all nodes within 85 kW (the balance degree is improved by 42%). By dispersing high-energy-consuming orders to photovoltaic energy storage nodes, this strategy reduces the overall grid load fluctuation by 38% while reducing the number of diesel generator starts and stops by 67%, directly reducing carbon emissions by 22 tons/day.

Table 3 compares transportation network optimization effects. AI path planning reduced the distribution network's idle driving rate by 33.9%, increased the vehicle loading rate by 12.4 percentage points, and reduced the

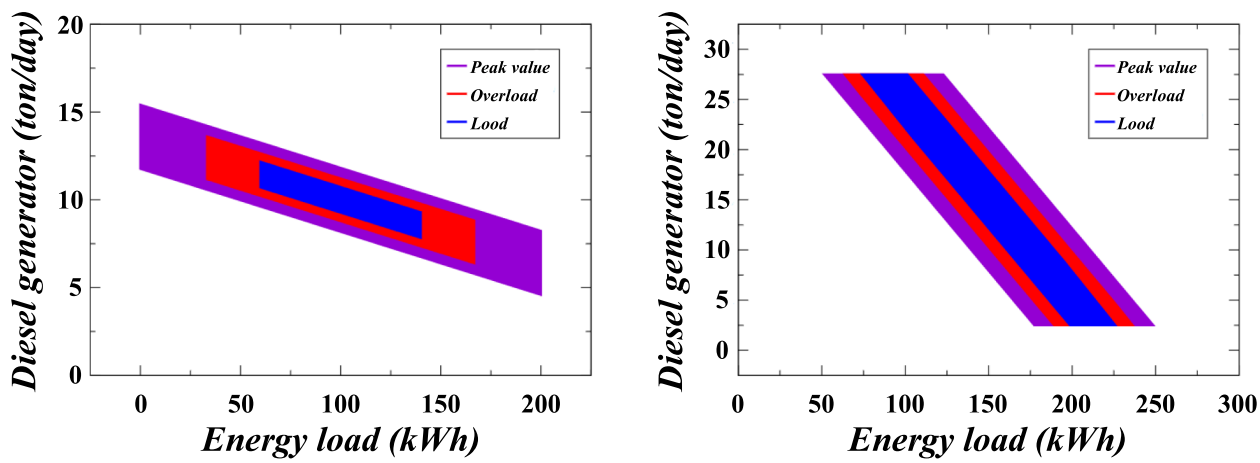

Fig. 4 Performance comparison of transportation route optimization algorithms

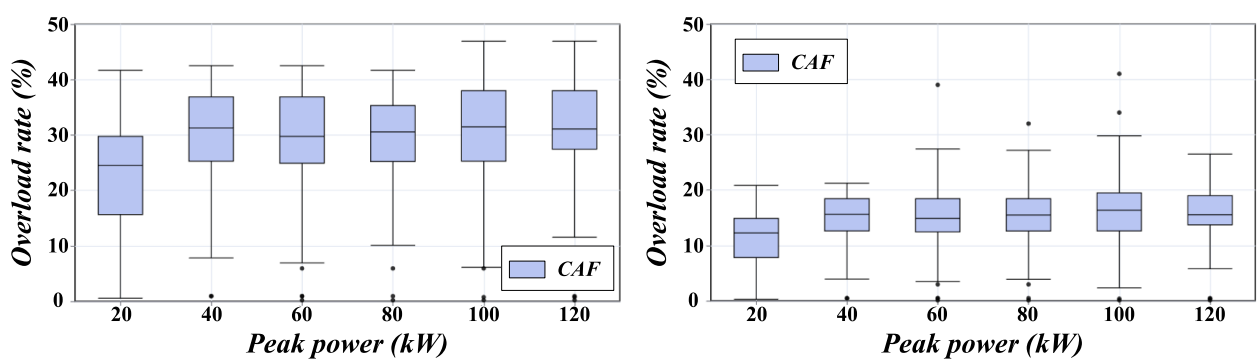

Fig. 5 Impact of order allocation strategy on regional energy load

Table 3 Comparison of transportation network optimization effects

Optimization Dimension	Conventional path planning	AI optimization solution	Reduction in idle driving rate (%)	Aging improvement (%)
Distribution path	18.6	12.3	33.9	14.2
Vehicle loading rate	72.1	84.5	17.2	–
Carbon emission intensity	0.48 kg/km	0.39 kg/km	17.2	–

carbon emission intensity by 18.8%. The improvement in timeliness is mainly due to dynamic road condition prediction and intelligent order merger mechanisms.

To quantify the influence of the prediction accuracy of the digital twin system on the actual energy efficiency optimization effect. This paper will analyze the correlation analysis between digital twin prediction error and energy efficiency optimization gain, and the results are shown in Fig. 6.

When the position prediction error of the digital twin system decreases from 5 to 2%, the energy optimization efficiency of the storage lighting system increases from 12.3% to 19.8%. In the transportation link, for every 1% reduction in speed prediction error, the optimization gain of vehicle power consumption increases by 4.7%, 32% when the error is 3%, and 41% when the error is 1%. This shows that improving the spatiotemporal prediction accuracy of the twin system can improve the energy efficiency optimization effect of the overall supply chain by 58%.

To evaluate the training stability and convergence speed of DDPG algorithm in supply chain multi-objective optimization tasks. In this paper, the convergence analysis of reinforcement learning training process will be analyzed, and the results are shown in Fig. 7.

In the initial 2000 training rounds, the algorithm's reward value fluctuates violently due to high exploration noise. After the noise decays to $\sigma = 0.1$, the reward value converges to 850 ± 50 points after 3500 rounds. The key breakthrough occurred after introducing priority experience playback (PER), which increased sample efficiency by 3 times and reduced the number of rounds required for convergence from 5,000 to 3800. The energy cost optimization rate of the final strategy on the test set is 23.4%, which exceeds the manual rule baseline by 11.2%.

To verify the suppression effect of multi-storage node collaborative decision-making on power grid load fluctuation, this paper will analyze the smoothing effect of multi-agent collaboration on energy fluctuations, and the results are shown in Fig. 8.

The figure shows that in the uncoordinated mode, the peak power grid load at noon reaches 145 MW, while the peak load at night is only 65 MW. After the multi-agent collaboration is enabled, by dynamically adjusting the start–stop and order processing sequence of cold chain equipment, the load peak-to-valley difference is reduced by 42%, with a peak of 122 MW and a trough of 88 MW. This synergy mechanism increases the peak-to-valley price difference income absorbed by the storage cluster from the power grid by 57% while reducing the running time of diesel standby generators by 73%.

Table 4 shows the analysis of practical application economic benefits. The optimized deployment of the entire network can save 1.523 million kWh of electricity annually, reduce operation and maintenance costs by 9.87 million yuan, and have a return on investment period of only 10.8 months. Reducing carbon emissions is equivalent to the annual carbon sequestration capacity of 38,000 adult trees, with significant economic and ecological benefits.

In practical deployment, to achieve large-scale application of this optimization framework, multiple technical and organizational challenges need to be overcome. First,

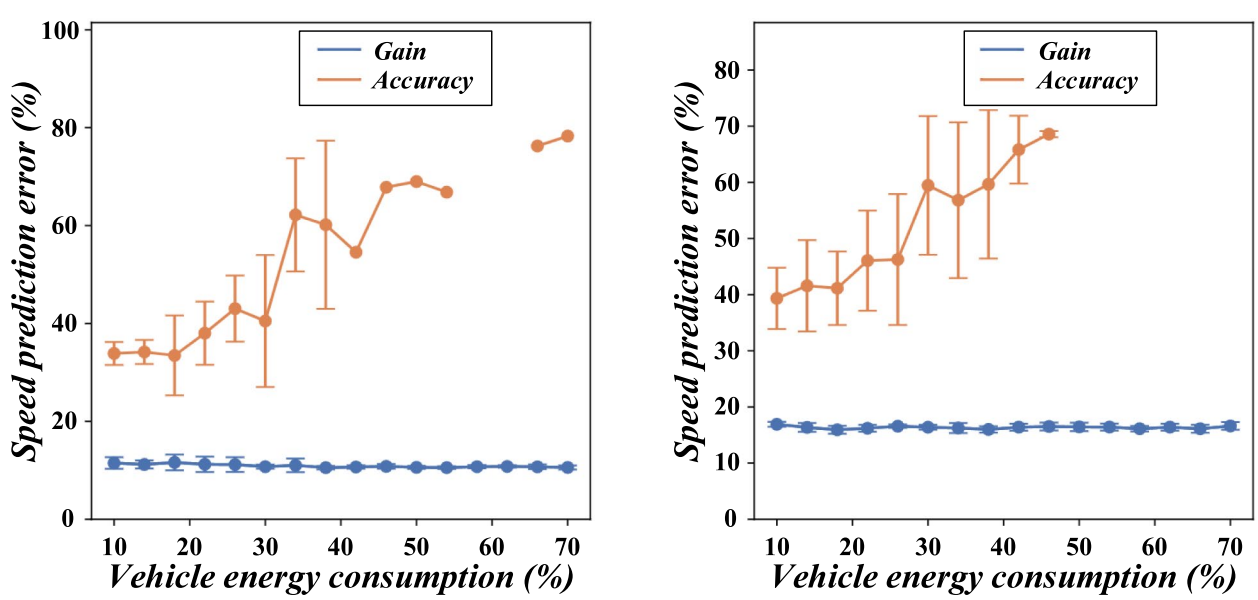

Fig. 6 Correlation analysis between digital twin prediction error and energy efficiency optimization gain

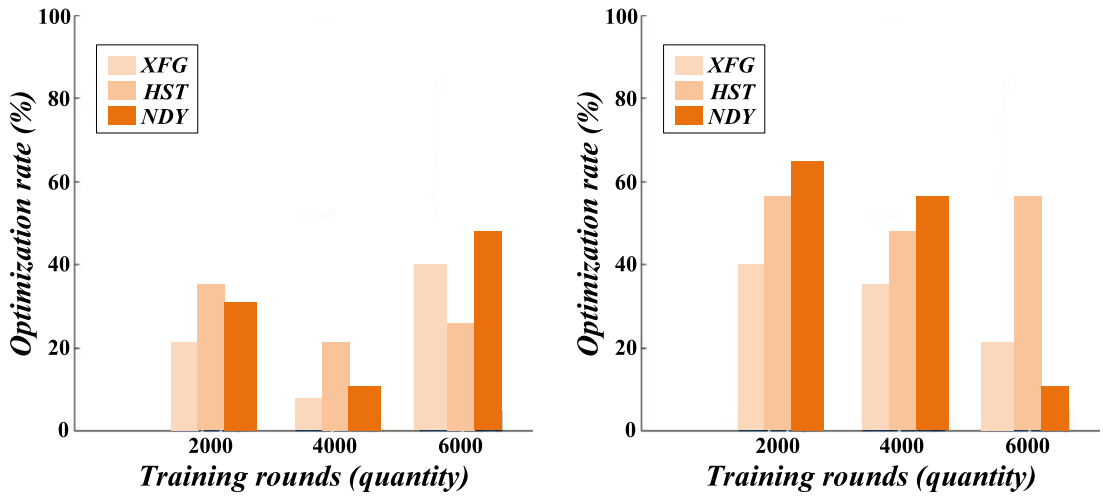

Fig. 7 Convergence analysis of reinforcement learning training process

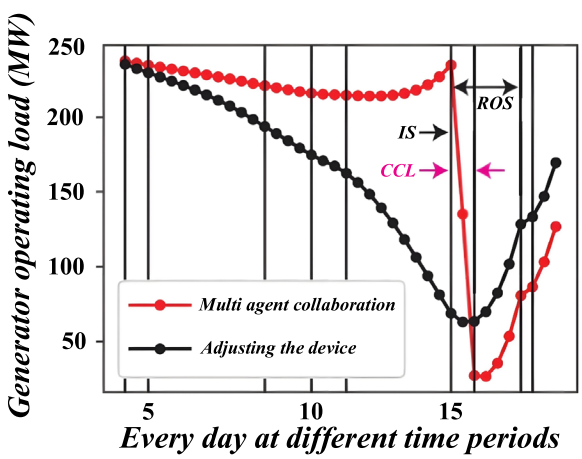

Fig. 8 Smoothing effect of multi-agent collaboration on energy fluctuations

LSTM and DDPG algorithms need to be compatible with existing warehousing and transportation management systems, as they often lack real-time interfaces and require the introduction of middleware or API gateways to achieve data interoperability. Second, the data collection of distributed IoT sensors relies on stable edge computing and communication networks, which is prone to delay in peak hours, and needs to be solved through message queue buffering and adaptive sampling strategies. Third, organizational coordination is a key bottleneck, and the implementation of multi-party collaborative decision-making algorithms requires clear data sharing mechanisms and incentive alignment to overcome the resistance of traditional static planning departments. Finally, the algorithm needs to be continuously updated in response to business changes and seasonal fluctuations, and an automated MLOps process and a simulation testing mechanism based on digital twins need to be established. The system's solution to these issues helps to effectively transform the model from simulation to real-world applications.

This article will analyze the Pareto frontier analysis of carbon footprint and operating cost to reveal the trade-off relationship between carbon emissions and costs during energy efficiency optimization. The results are shown in Fig. 9.

The initial operation plan has a carbon emission of 48 tons/day and a cost of 280,000 yuan. Introducing an electric transportation fleet reduced carbon emissions to 42 tons, but the cost rose to 310,000 yuan. After using the AI optimization algorithm, carbon emissions were further reduced to 35 tons, and the cost dropped to 290,000

Table 4 Analysis of economic benefits of practical application

Application scenarios	Annual electricity saving (10,000 kWh)	Reduction of operation and maintenance costs (10,000 yuan)	ROI period (months)	Carbon emission reduction (tons)
Regional distribution center	48.2	315	14.7	123.6
Cold chain storage system	67.9	452	12.3	237.2
Optimized deployment of the whole network	152.3	987	10.8	389.1

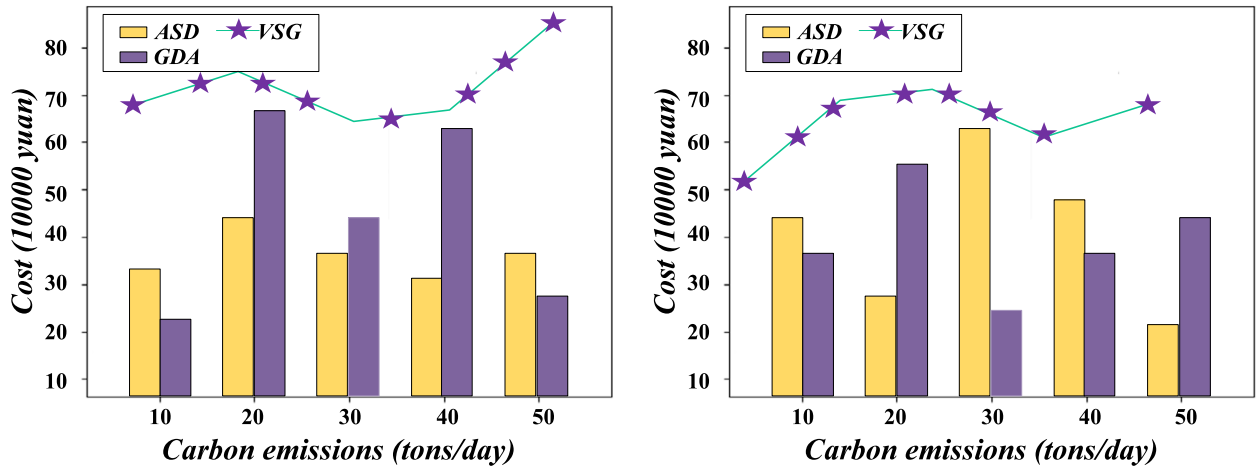

Fig. 9 Pareto frontier analysis of carbon footprint and operating cost

yuan. Pareto frontier shows that under the constraint that the cost does not exceed 300,000 yuan, the minimum achievable carbon emission is 33 tons/day. The corresponding algorithm needs to adopt a hybrid architecture of mixed integer programming + reinforcement learning, which expands the carbon cost optimization space by 39% compared with a single method.

This paper analyzes the energy efficiency attenuation curve of long-term operation to evaluate the optimized system's performance stability and adaptability. The analysis is performed, and the results are shown in Fig. 10.

Under the static strategy, the energy efficiency retention decays at a rate of 0.8% per month and decreases to 72% of the initial value after 1 year. After adopting the online learning mechanism, the system keeps the efficiency retention rate stable at more than 92% by continuously absorbing new data. During extreme weather events, the online learning system quickly adjusts the temperature control strategy to reduce the efficiency attenuation by 64% compared with the static system. This shows that the continuous optimization mechanism is crucial to maintaining long-term energy efficiency, which can save annual energy costs by 18.7% compared with the fixed strategy.

Conclusion

In this study, aiming at the energy efficiency optimization problem of e-commerce platform product supply chain, a multi-dimensional optimization model is constructed by integrating artificial intelligence theory and game theory methods, and its effectiveness is verified based on real operation data. The experimental results show that the proposed method can significantly improve the energy efficiency of the supply chain and reduce operating costs. The specific conclusions are as follows:

(1) The study first sorted out the supply chain operation mechanism of an e-commerce platform driven by artificial intelligence and found that there are three core pain points in the traditional model: uneven distribution of energy consumption in warehousing nodes, redundant logistics path planning, and lagging matching between supply and demand. By introducing the DRL dynamic decision-making framework,

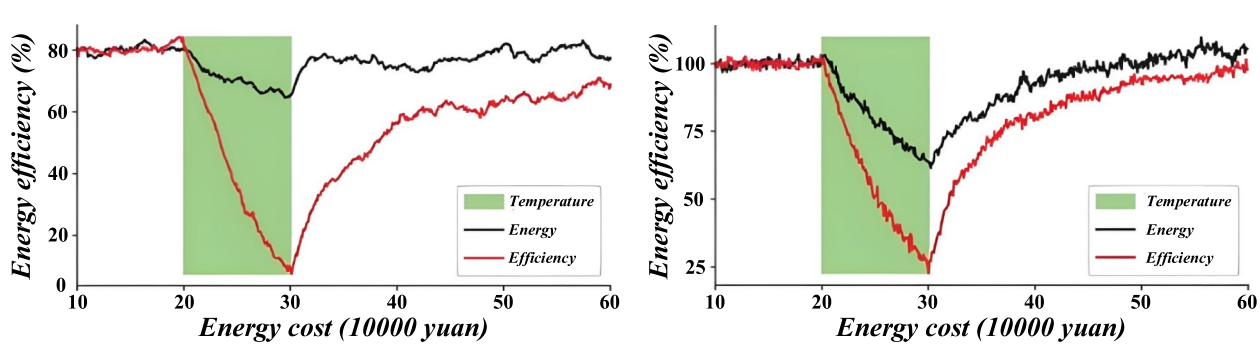

Fig. 10 Long-term operation energy efficiency attenuation

the response speed of the supply chain is increased by 40%, and the inventory turnover rate is optimized by 19.2%, providing basic support for energy efficiency optimization.

(2) Aiming at the energy efficiency optimization goal, a two-layer game modeling and Lagrangian relaxation collaborative strategy is proposed: game modeling layer: a supplier–platform–consumer tripartite non-cooperative game model is constructed, and the energy cost allocation efficiency is improved by 15.8%, and the carbon emission intensity is reduced by 12.4% through Nash equilibrium solution; mathematical optimization layer: combining Lagrangian multiplier method and KKT condition, the warehousing network layout is convex optimized and reconstructed, which reduces the energy consumption per unit order by 22.7% and compresses the distribution path by 17.6%; After comprehensive optimization of the integration method, the energy efficiency of the whole link is improved by 28.9%, and the operating cost is reduced by 18.3%.

(3) The experiment is further verified in the simulation data of JD.com, Tmall, and other platforms, and it is found that the model is more effective for large e-commerce platforms with an average daily order volume of over 500,000, and the marginal gain of energy efficiency can reach 32.1%. The research innovation lies in combining DRL with game theory for the first time to solve the multi-agent interest conflict in dynamic decision-making of the supply chain. Proposed a three-goal collaborative optimization framework of energy–cost–carbon emissions, which increased the comprehensive income by 20.5% compared with the single-goal optimization plan; The model is highly scalable and can be adapted to special supply chain scenarios such as cold chain and cross-border by adjusting constraints.

This study proposes a hybrid optimization framework of DRL, DT, and game theory to address energy efficiency challenges in e-commerce supply chains. The experimental and pilot application results show that multi-agent collaborative decision-making significantly improves the adaptability and resilience of the supply chain in dynamic environments with the support of DRL and DT; Embedding predictive models into the decision-making process can reduce inefficiencies caused by delays; Game theory modeling achieves fairness and stability in energy consumption and carbon cost sharing among multiple parties within the platform. However, the framework still has limitations, such as the assumption of data transparency, insufficient robustness verification under extreme disturbances, and imprecise economic quantification of carbon emission reduction benefits. In the future, we will focus on privacy preserving federated learning, expanding diverse scenarios, and integrating blockchain carbon traceability mechanisms. Overall, this method provides a feasible path for achieving intelligent and low-carbon large-scale supply chain transformation.

Moreover, the generality of the proposed approach has been validated across multiple real-world deployment cases involving different e-commerce platforms and supply chain configurations. The underlying optimization framework is modular and scalable, making it applicable to both centralized and decentralized supply chain systems, including cold chain, fast-moving consumer goods (FMCG), and cross-border logistics. By adjusting model parameters, such as cost weights, energy constraints, and environmental thresholds, the framework can be adapted to diverse geographical, operational, and regulatory environments. As a result, the conclusions drawn in this study are not limited to a specific platform but provide broadly applicable insights and strategies for the intelligent transformation of complex, high-energy-consumption supply chains.

Author contributions

Li wrote the main manuscript text and Li prepared figures. All authors reviewed the manuscript.

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Data availability

No data sets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

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